THE KERNEL OF ANY SURJECTION FROM A POLYNOMIAL RING ONTO A FINITELY PRESENTED ALGEBRA IS FINITELY GENERATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- A presentation of a finitely presented algebra. Used at the first line of the counted proof (line 790). Supplies finitely many variables y_j and finitely many relations f_l for S over R.
- Comparison of two presentations through an auxiliary polynomial ring. Used in the middle of the counted proof (lines 795-797). Lets the kernel of the given surjection be described as the image of an ideal under a map that fixes the x_i.

Lemma 0.1. *Let $R \to S$ be a ring map of finite presentation. For any surjection $\alpha : R[x_1, \ldots, x_n] \to S$ the kernel of α is a finitely generated ideal in $R[x_1, \ldots, x_n]$.*

Proof. Write $S = R[y_1, \ldots, y_m]/(f_1, \ldots, f_k)$. Choose $g_i \in R[y_1, \ldots, y_m]$ which are lifts of $\alpha(x_i)$. Then we see that $S = R[x_i, y_j]/(f_l, x_i - g_i)$. Choose $h_j \in R[x_1, \ldots, x_n]$ such that $\alpha(h_j)$ corresponds to $y_j \bmod (f_1, \ldots, f_k)$. Consider the map $\psi : R[x_i, y_j] \to R[x_i]$, $x_i \mapsto x_i$, $y_j \mapsto h_j$. Then the kernel of α is the image of $(f_l, x_i - g_i)$ under ψ and we win. □

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